

Dhruv Samdani

Software Engineer • Backend & Distributed Systems • Machine Learning

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LinkedIn • GitHub • LeetCode • CodeChef • Codeforces

EDUCATION

Indian Institute of Information Technology, Allahabad

B.Tech in Information Technology | CGPA: 8.64 / 10.00

Aug 2023 – May 2027

Prayagraj, UP

Green Field Public School, Manasa

Class XII, CBSE | 96.0%

Apr 2020 – May 2022

Madhya Pradesh

TECHNICAL SKILLS

Languages	C++, C, Java, Python, JavaScript / TypeScript, SQL
Backend & Systems	Node.js, Express.js, FastAPI, Crow (C++), REST API Design, Multithreading & Concurrency, OOP / LLD
CS Fundamentals	Data Structures & Algorithms, System Design, Operating Systems, Computer Networks, DBMS
Databases	PostgreSQL, MySQL, MongoDB, Redis, Database Indexing & Caching
ML & AI	PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, NLP, Reinforcement Learning
Cloud & DevOps	AWS (EC2, S3, RDS), Docker, Kubernetes, Apache Kafka, CI/CD, Git, Linux

EXPERIENCE

Full-Stack Development Intern

ADY Infratech

Apr 2026 – Jun 2026

- Engineered a real-time ERP system for 2 mixing plants, raising vehicle-tracking accuracy from **70% to 90%**.
- Built order and billing modules processing 1,200+ invoices/week **30% faster**, cutting billing errors by 25%.
- Designed an internal reconciliation tool that reduced manual data-entry errors by **30%**, now the firm's official workflow.

PROJECTS

DynamicKV — Scalable NoSQL Key-Value Store

C++17, Crow, Robin-Hood Hashing, Bloom Filters, REST API

GitHub

- Built a NoSQL engine in modern **C++17** with a custom **Robin-Hood hashing** HashMap for fast in-memory lookups.
- Designed a **segmented on-disk storage** layer with per-segment **Bloom filters** and tunable sizing.
- Exposed a **thread-safe REST API** over HTTP via the Crow framework for concurrent reads and writes.

Historical Entity Linking — Multilingual NER & Disambiguation

Python, PyTorch, XGBoost, BLINK, mReFinED, Transformers

GitHub

- Reproduced **BLINK** and **mReFinED** baselines, benchmarking multilingual linking on MEWSLI-9 across 9 languages.
- Engineered MHEL-LLaMo, an XGBoost confidence router reaching **82% routing accuracy** (+12% over baseline).
- Ran systematic FP/FN error analysis and offset-validation fixes, improving single-GPU reproducibility.

NyayaSetu — AI-Powered Legal Intelligence Platform

Next.js, React, TypeScript, Gemini 2.0 Flash, Speech-to-Text

GitHub

- Built a legal-research assistant surfacing the **5 most relevant Supreme Court precedents** per query.
- Developed a judgment predictor plus NyayKaksh, a real-time **English-to-Hindi speech translator** with per-sentence AI analysis.
- Integrated **Google Gemini 2.0 Flash** for case analysis and follow-up Q&A behind a Next.js + TypeScript interface.

InvestorPulse — Stock Performance & Recommendation Platform

Next.js 14, FastAPI, PostgreSQL, Redis, Docker, Yahoo Finance API

GitHub

- Built a **full-stack fintech platform** serving explainable, risk-labelled stock recommendations to retail investors.
- Architected a **FastAPI** backend with Yahoo Finance integration and a Next.js dashboard, containerized with **Docker**, Postgres, and Redis.
- Implemented a 5-level risk model with **SHAP-based explanations**, deployed to Vercel and Render.

ACHIEVEMENTS & LEADERSHIP

- Competitive Programming:** LeetCode Knight (peak 1880) — **237th of 28,244** in Weekly Contest 455 (top 1%); **618th of 18,797** in CodeChef START193D.
- Hackathons:** **134th** in Tata AI Challenge 2026; **Semifinalist**, Flipkart GRID 7.0.
- Leadership:** Coordinator, Effervescence'25 (IIIT-A cultural festival); Winner, IIITAMUN'23 and Bharat Ko Jano National Quiz.
- Certifications:** Oracle OCI AI Foundations Associate; Oracle OCI Foundations Associate.